StructEst Pro — CAD Drawing Standard for Automatic Take-Off

Follow this standard and StructEst will extract walls, columns, beams, slabs, doors, windows and sanitary fixtures **automatically** — no manual measurement. If you only have DWG, convert it to DXF first (AutoCAD "Save As → DXF", or a free ODA File Converter). DWG cannot be read directly.

1. Units — author in METERS

- Set the drawing unit to **meters** and set the DXF header variable `$INSUNITS = 6`. StructEst reads this to know the drawing scale.
- If `$INSUNITS` is missing, StructEst auto-guesses the unit from the drawing size (mm / cm / m / ft) and flags it as "auto-guessed" — verify it in-app.
- Draw geometry at true size (e.g. a 9 m wall is 9.0 units long).
- Do **not** draw 1 unit = 1 mm or 1 unit = 1 inch — quantities will be wrong.
- If you cannot set meters, set a known scale and calibrate manually in-app.

2. Layer standard

Put each component type on its recommended layer. Layer names are matched case-insensitively and also by keyword, so `A-WALL-EXT` and `walls_outer` both work — but the exact names below extract with zero guessing.

Layer	Purpose	How it is measured
`A-WALL-EXT`	External brick/block walls (centerline)	Length → m (× thickness × height = m^3)
`A-WALL-INT`	Internal partition walls (centerline)	Length → m
`A-DOOR`	Door leaf symbols as INSERT blocks (DOOR_*)	Count → pcs (deducts area in walls/plaster)
`A-WIND`	Window symbols as INSERT blocks (WIN_*)	Count → pcs (deducts area)
`S-COLS`	RCC columns (closed LWPOLYLINE or COL_* block)	Area → m^2 (× height = m^3), or count
`S-BEAM`	Beam centerlines (LINE / LWPOLYLINE)	Length → m (× section = m^3)
`S-SLAB`	Slab outline (closed LWPOLYLINE)	Area → m^2 (× thickness = m^3)
`P-SANTRY`	Sanitary fixtures: WC, basin, sink (INSERT blocks)	Count → pcs
`A-ROOM`	Room name tags (TEXT / MTEXT)	Labels (drives area schedule)
`A-GRID`	Column grid lines	Reference only
`A-FURN`	Furniture & fittings	Excluded from BOQ
`A-ANNO-TEXT`	Notes & title block	Excluded from BOQ
`A-ANNO-DIMS`	Dimension entities	Excluded from BOQ

Layer	Purpose	How it is measured
`DEFPOINTS`	CAD construction points	Excluded from BOQ

3. Block symbol naming (doors, windows, columns, fixtures)

Doors, windows, columns and sanitary fixtures should be **INSERT block symbols**, not loose geometry. Block names are the strongest signal StructEst reads, so name them with a component prefix:

- DOOR_900 / DOOR_750 / DOOR_3-7 → door leaf symbol (swing arc + leaf lines)
- WIN_1200 / WIN_4X5 → window symbol (frame rectangle + mullions)
- COL_300X450 / PILLAR_1 → column symbol (footprint cross-section)
- WC / SINK / BASIN / SHOWER → sanitary fixture symbol
- Prefix every symbol block with the component type so it extracts by name.

If a symbol is left as an anonymous block, StructEst falls back to a geometric heuristic (an ARC + lines = door; a plain rectangle = window), but explicit names are far more reliable.

4. Supported geometry — curves are measured exactly

StructEst flattens every entity to true geometry before measuring, so these all contribute correct quantities:

- **LINE / LWPOLYLINE / POLYLINE** — including polyline **bulges**: arcs between vertices are measured along the curve, never the straight chord.
- **ARC / CIRCLE** — measured as radius × sweep angle (a quarter-circle wall bay is its true arc length; round columns use the full circumference).
- **ELLIPSE / SPLINE** — tessellated and measured along the curve.
- **INSERT blocks** — block contents are flattened with their rotation and scale, so nested symbols render in the preview and their geometry can be measured.
- Annotation noise (dimensions, hatches, DEFPOINTS, title-block layers) is hidden by default — toggle any layer back on from the layer list.

5. What each component becomes in the BOQ

Component	BOQ label	Category	Measure
wall	Brick Masonry / Wall	Superstructure	length
column	RCC Column	Superstructure	area
beam	RCC Beam	Superstructure	length
slab	RCC Slab	Superstructure	area
window	Window	Finishing	count
door	Door	Finishing	count
toilet	Sanitary Fixture (WC)	MEP Works	count
fixture	Sanitary Fitting (basin/sink)	MEP Works	count

- **length** measures (walls, beams) → run length in m; × thickness × height = m^3.
- **area** measures (columns, slab) → footprint/surface area in m^2; × height/thickness = m^3.
- **count** measures (doors, windows, fixtures) → number of symbols found (pcs).

6. Quick checklist before exporting

- ☐ `$INSUNITS = 6` (meters) set in the header.
- ☐ Walls drawn as centerline LINE / LWPOLYLINE on a `*WALL*` layer.
- ☐ Columns as a closed outline (or a `COL_*` block) on a `*COL*` layer.
- ☐ Beams as centerlines on a `*BEAM*` layer; slab as a closed outline on a `*SLAB*` layer.
- ☐ Doors/windows as INSERT blocks named `DOOR_*` / `WIN_*`.
- ☐ Fixtures as INSERT blocks named `WC` / `SINK` / `BASIN`.
- ☐ Saved as **DXF** (not DWG).

7. Non-standard drawings — AI Layer Mapper

Drawings that do not follow this standard still extract. Import the DXF, open the AutoCAD Vector tab and run the **AI Layer Mapper**: Gemini proposes layer and block renames toward the standard from a geometry fingerprint (entity mix, lengths, closed areas, text tags — never the drawing geometry itself). You review every row and apply the ones you accept; annotations can be marked "ignore". Renames rewrite names only — geometry is never changed — the app re-extracts immediately, and the normalized DXF is available for download so what you measured is exactly what you keep.

Generated by StructEst Pro from the live DXF standards (DXF_STANDARD_LAYERS, DXF_BLOCK_STANDARD, DEFAULT_RULES) — the document cannot drift from what the extractor actually reads.